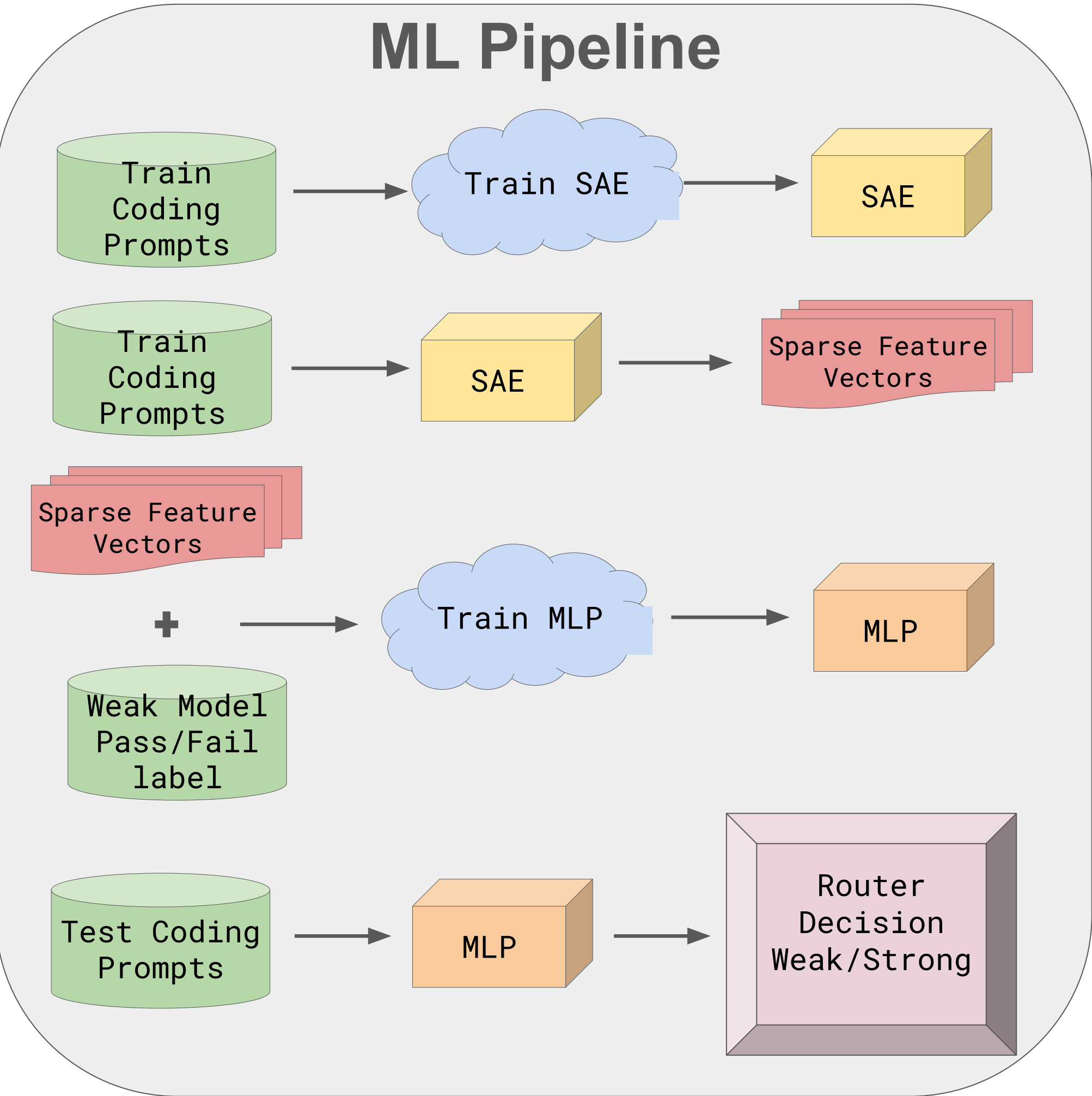


Purpose
Current routers use black box embeddings that do not provide any information regarding why a model was routed to.

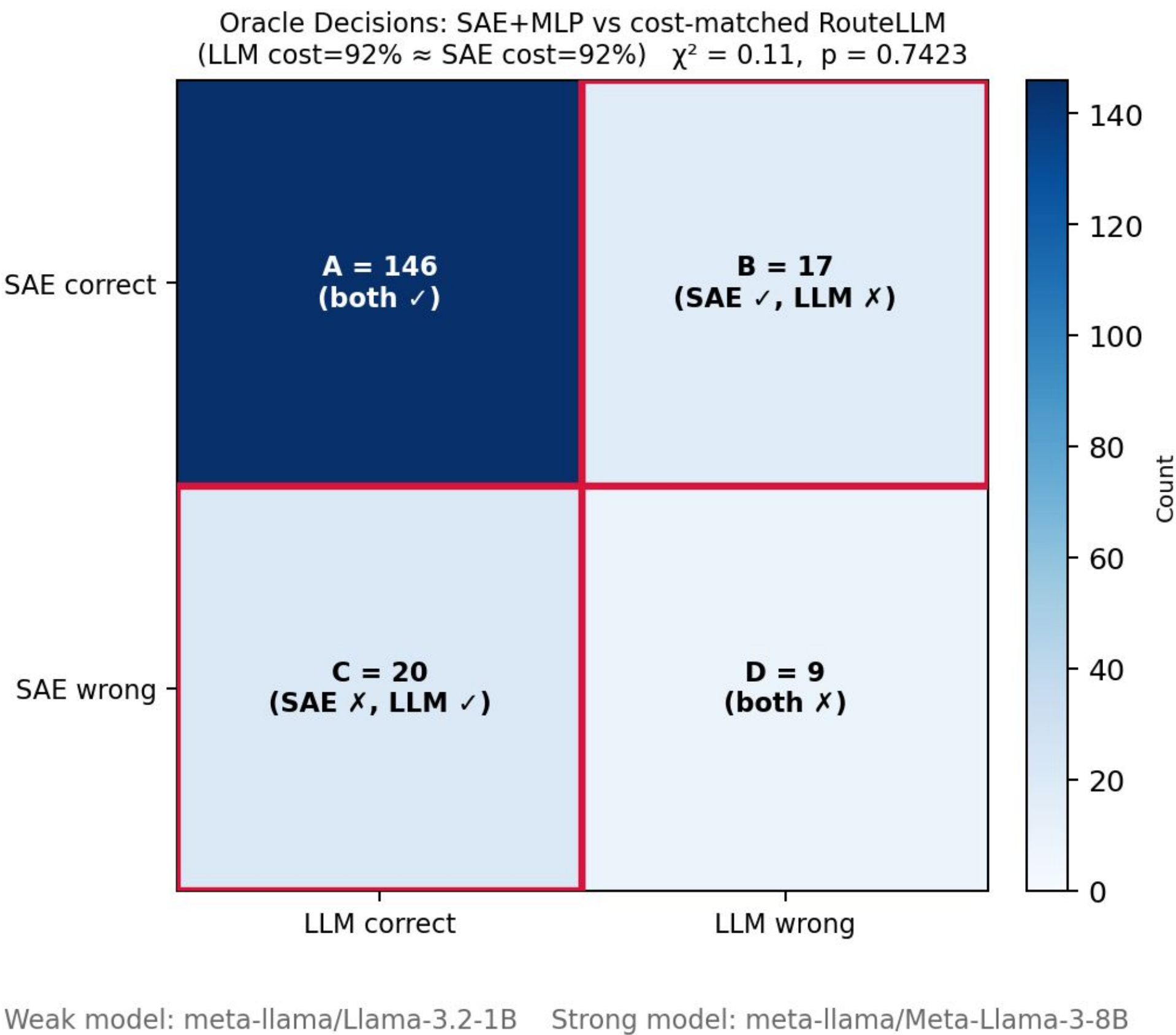
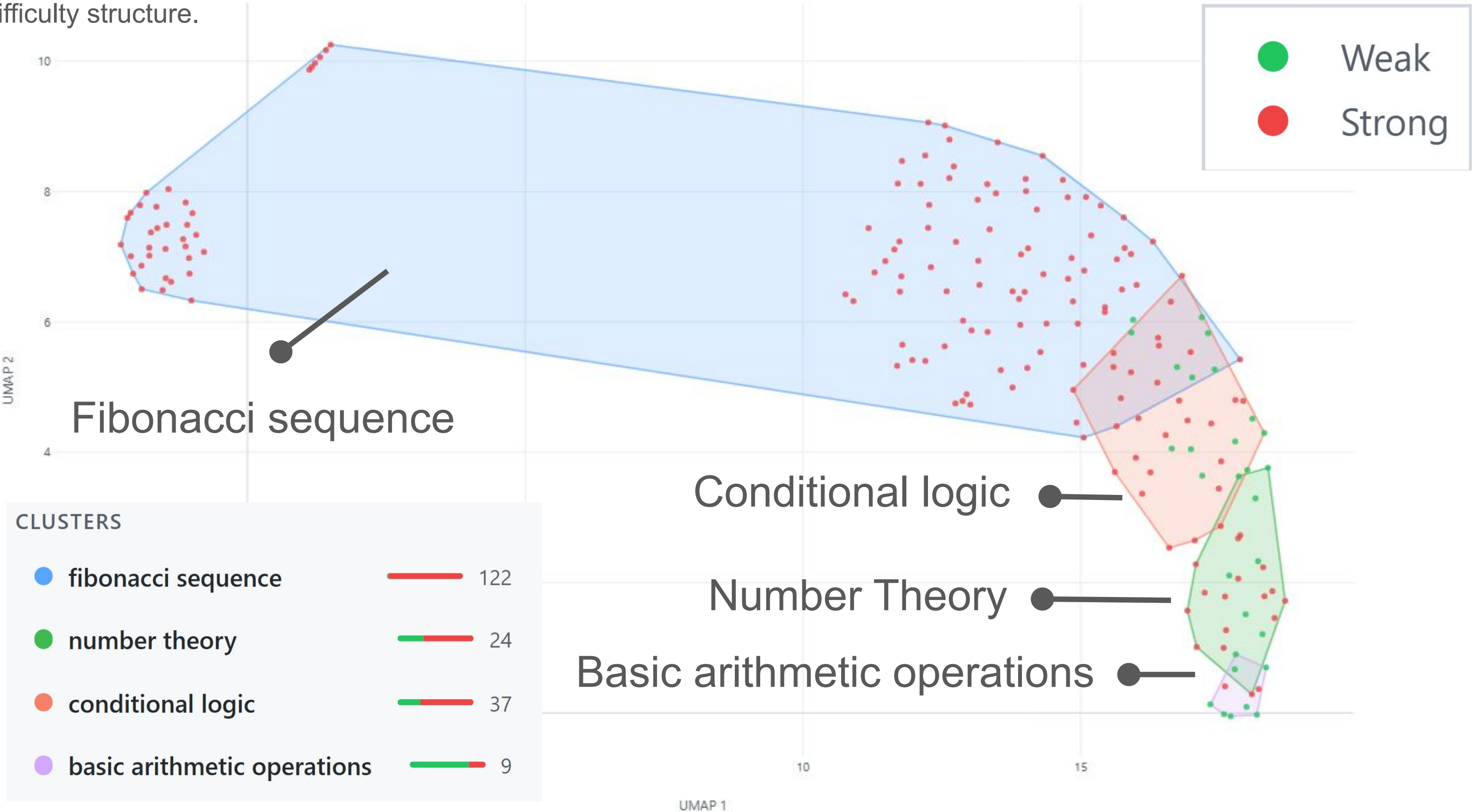
Hypothesis
Sparse feature vectors provide sufficient information on **coding** related prompts to make routing decisions.

Baseline
Route LLM’s Toughness Threshold based router is our baseline. It makes use of a BERT encoder and a Route LLM trained MLP to output prompt toughness. Knee method used to find ideal threshold for our train set.

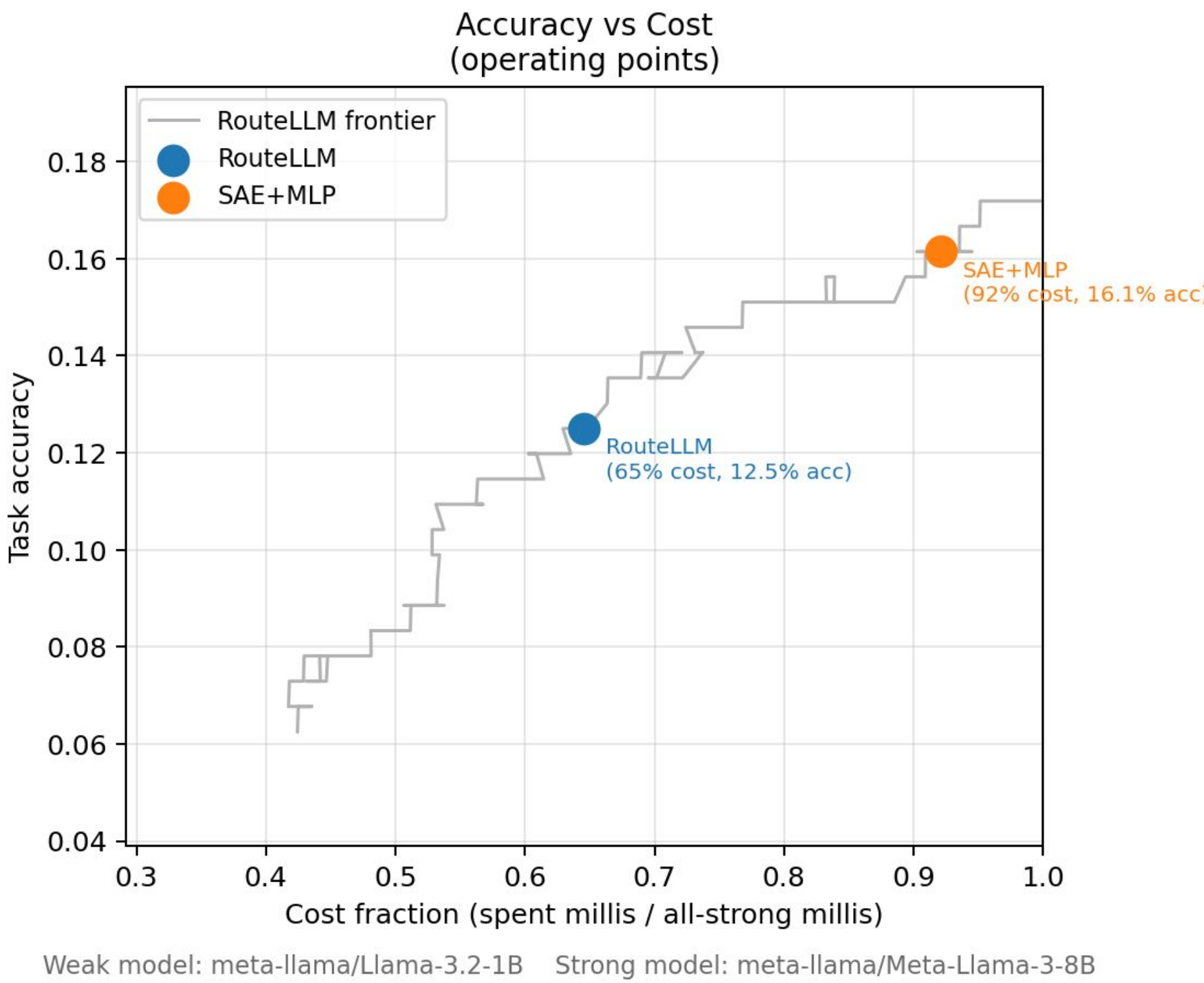


 **Weak Model:** meta-llama/Llama-3.2-1B
Strong Model: meta-llama/Meta-Llama-3-8B

Clusters of sparse feature vectors reveal that weak model queries group together in the bottom right while harder problems dominate the large blue region, showing that SAE features capture routing relevant difficulty structure.



The Results above and below show that our model has hit the same frontier in the dataset that the Route LLM router hit from cost (run-millis) across from accuracy.



 **LLM Clustering Labeler:** OpenAI/GPT-4o-Mini